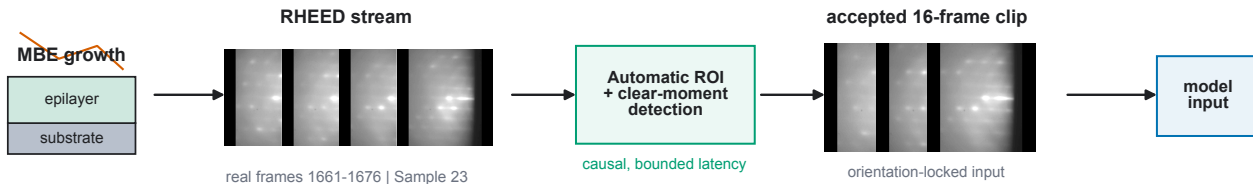


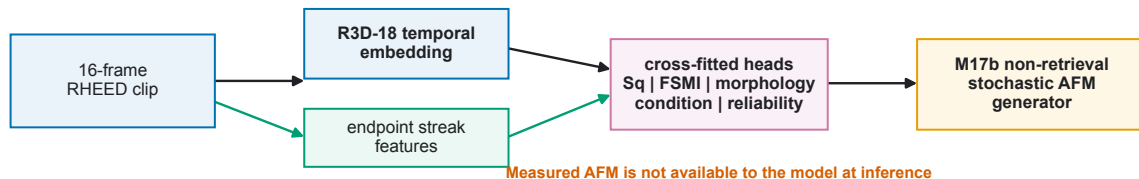
a

In-situ acquisition and causal event selection



b

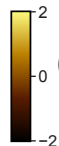
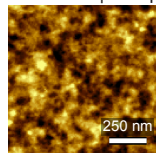
Hybrid physics-AI inference



c

Quantitative output

generated AFM | Sample 23



Sq

0.83 nm

FSMI

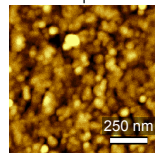
1.29 nm

reliability

71 / 100

one physically scaled realization; ensemble retained for uncertainty analysis

measured AFM | evaluation only



not an inference input